

Bayesian Estimation of mmt8b0ccBin3_covid_bT_baa: Methodology, Convergence, and Determinacy/E-Stability Results

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Abstract

This note documents the Bayesian estimation of an extended version of the shopping-time/banking-sector model with monetary and fiscal policy interactions, adding (i) an endogenous debt-to-GDP ratio observable built from FRED federal debt held by the public and nominal GDP, (ii) a Baa corporate bond yield in place of the bank-prime rate for the loan-rate observable, (iii) a genuine OccBin piecewise-linear zero-lower-bound (ZLB) likelihood in place of a linear-Kalman-filter approximation, and (iv) a single independent, identically distributed (iid) shock to the labor-leisure margin standing in for the COVID-19 labor-supply collapse. This round additionally recalibrates two structural parameters – the steady-state tax target $\bar{\tau}$ (from 0.20 to 0.14, implying a 6%-of-GDP steady-state deficit, more realistic than the previous $\approx 0\%$ balanced-budget target) and the shopping-time curvature χ (from 1.0 to 0.5, chosen after comparative-statics checks showed it substantially lowers steady-state inflation via higher money demand) – both of which shift the model’s steady state and require re-running mode-finding and estimation from scratch. We report the two-stage mode-finding and jump-scale tuning – including a switch from a diagonal (`prior_variance`) to a Hessian-based MCMC jumping covariance, adopted in an earlier round to resolve a chain-disagreement problem that produced bimodal-looking pooled posterior densities, and a further switch to a jumping covariance built directly from a preliminary round’s own pooled posterior draws once the Hessian-based approach itself began showing renewed (if milder) chain disagreement under the recalibrated model – used to obtain a well-mixing Markov chain Monte Carlo (MCMC) sample under OccBin’s nonsmooth likelihood, the resulting posterior distribution and Geweke (1992) convergence diagnostics, a determinacy and expectational-stability (E-stability) map for the non-ZLB (RELAX) regime evaluated at the posterior means, and a table of annualized steady-state rates implied by those estimates.

This revision reports a further re-estimation round on top of the above, prompted by two changes: (i) a corrected household budget-constraint identity for reserves n_t , which previously used a fixed steady-state placeholder for the debt terms rather than the genuine (lagged and current) debt state, and (ii) substantially less informative priors on the Taylor-rule policy-response coefficients b_π and b_y (Gamma(1.00, 1.00) and Gamma(1.00, 0.70), replacing a tight Normal(2.10, 0.15) and Gamma(0.28, 0.10) respectively) together with a lower prior mean for the fiscal-feedback parameter φ_b (0.05, from 0.10), so that the posterior on the inflation-response coefficient in particular reflects what the data support rather than a tight prior. All posterior, convergence, determinacy, and steady-state results reported below are from this latest round; the tuning history of the earlier (tight-prior) round is retained for context.

1 Model Extensions

The estimated model, `mmt8b0ccBin3_covid_bT_baa.mod`, extends the baseline quarterly `mmt8b0ccBin3` specification along four dimensions.

Debt-to-GDP observable. An endogenous debt-to-GDP ratio is added as

$$bT_t = \frac{b_t}{1 + g},$$

where b_t is the model's real net-debt-to-output ratio and g is steady-state government spending as a share of output (so $1 + g$ approximates steady-state GDP in the model's units). The corresponding observable is

$$bT_t^{obs} = \log(bT_t) - \log(\overline{bT}) + \varepsilon_{me,t}^{bT},$$

built from FRED series `FYGFDPUN` (federal debt *held by the public*, not gross/total public debt) divided by GDP (nominal GDP), log-demeaned over the sample. An earlier version of this observable used `GFDEBTN` (total public debt) and demeaned in levels rather than logs, inconsistent with the measurement equation above; both issues are fixed in the current data build (see Section 2.1).

Baa-based loan rate. The loan-rate observable r_t^{obs} is constructed from Moody's Seasoned Baa Corporate Bond Yield (FRED series `BAA`) rather than the bank prime rate (`MPRIME`) used in earlier versions of this model. The Baa series carries genuine credit-spread variation (widening in recessions, narrowing in recoveries) instead of being mechanically pinned to the federal funds rate plus a fixed spread, and its 2009–2025 sample average sits closer to the model's steady-state loan rate.

Genuine OccBin ZLB likelihood. The interest-rate rule on reserves is subject to a zero lower bound,

$$r_t^n = \max \left[1, (r_t^\ell)^{a_i} \left(\frac{\pi_t}{\pi} \right)^{b_\pi} \left(\frac{y_t}{y} \right)^{b_y} e^{z_t^r} \right],$$

implemented via Dynare's `occbin_constraints` block. Earlier versions of this file carried the `occbin_constraints` specification syntactically but disabled OccBin's piecewise-linear Kalman filter and smoother at estimation time (`options_.occbin.likelihood.status = false`), falling back to a standard linear filter that ignores the bound. This estimation instead sets `options_.occbin.likelihood = true` together with `options_.occbin.smoother.periodic_solution = true` and `options_.occbin.regular = 1e-3`, so the likelihood genuinely enforces the ZLB across both binding episodes in the sample (2009Q1–2015Q4 and 2020Q1–2021Q4).

COVID labor-supply shock. Rather than reproduce the multi-shock, regime-switching COVID mechanism of Cardani, Pfeiffer, Ratto, and Vogel (2023), the model adds a single iid shock ε_t^L to the household labor-leisure first-order condition,

$$\frac{c_t}{z} h_t^\psi + \chi h_t = \frac{1}{\eta} \exp(\varepsilon_t^L),$$

with $\varepsilon_t^L \sim N(0, \sigma_L^2)$ and σ_L estimated. A negative realization of ε_t^L shrinks the right-hand side, forcing lower labor supply for given consumption and wages – a negative labor-supply

shock. A deterministic one-period dummy (`varexo_det`) was considered first, matching a specific calendar quarter, but Dynare 7.1's estimation routine (`dynare_estimation_init.m`) does not read `varexo_det` variables from the estimation data file, so such a dummy would be silently ignored by the Kalman filter. The iid `varexo` shock avoids this and lets the smoother (when it converges) attribute whatever realization the data require to the appropriate quarter without a hardcoded date.

Recalibration: $\bar{\tau}$ and χ . Two calibrated (not estimated) structural parameters were revised this round. $\bar{\tau}$: the steady-state tax target, entering the tax rule

$$\tau_t = \tau_{t-1}^{\rho_\tau} \bar{\tau}^{1-\rho_\tau} \left(\frac{b_{t-1}}{(1-\omega)b_0} \right)^{\varphi_b} e^{\varepsilon_t^\tau},$$

was lowered from 0.20 to 0.14. Because government spending is fixed at $g = 0.2$, the steady state satisfies $\tau_{ss} = \bar{\tau}$ exactly and $b_{ss} = b_0$ exactly (verified numerically), so the steady-state deficit $g - \bar{\tau}$ moved from $\approx 0\%$ to 6% of GDP – a more realistic target for the 2009–2025 sample away from the ZLB than a balanced budget. Comparative statics at the previous round's posterior means showed this relationship is steep: at $\bar{\tau} = 0.15$, steady-state annual inflation was 8.66%; at $\bar{\tau} = 0.12$, 15.85%; at $\bar{\tau} = 0.05$ the steady-state solver failed to converge at all, consistent with the model approaching the peak of its seigniorage (Laffer-type) curve. χ : the shopping-time curvature parameter in $h_{st} = (1/\chi)(v_{a,t}c_t/d_t)^\chi$, was lowered from 1.0 to 0.5. A comparative-statics sweep ($\chi \in [0.3, 2.0]$, other parameters at the previous round's posterior means) showed a smooth, monotonic relationship between χ and steady-state inflation – lower χ raises real money demand substantially (roughly 8-fold from $\chi = 2.0$ to $\chi = 0.3$ in that sweep), which lowers the inflation tax needed to finance a given deficit. Both parameters remain calibrated rather than added to `estimated_params`.

Reserve equation fix: genuine debt-state coupling. The equation linking reserves n_t to currency and debt (the model's analog of the household budget-constraint identity $n_t = m_{t-1}/\Pi_t + b_{t-1}/\Pi_t - b_t/r_t$) previously read

$$n_t = \frac{m_{t-1}}{\pi_t} + b_0(1-\omega) \left(\frac{1}{\pi_t} - \frac{1}{r_t^\ell} \right),$$

using the fixed steady-state debt target $b_0(1-\omega)$ in place of the actual (lagged and current) debt state – meaning n_t never responded to fluctuations in outstanding debt at all. This is corrected to

$$n_t = \frac{m_{t-1}}{\pi_t} + \frac{b_{t-1}}{\pi_t} - \frac{b_t}{r_t^\ell},$$

the model's direct analog of the theoretical identity. This is safe for Blanchard-Kahn purposes because b_t 's own law of motion (the debt-issuance equation, which fixes the r_t^ℓ/π_t ratio at its steady-state value $r^{\ell,ss}/\pi^{ss}$ for exactly this reason) never depends on any forward-looking variable, so it remains a purely backward/exogenous recursion regardless of where else it appears contemporaneously. Verified directly: steady-state values and the Blanchard-Kahn eigenvalue count (4 forward-looking variables, order and rank conditions satisfied) are unchanged before and after the fix.

Less informative Taylor-rule priors. The priors on the inflation- and output-response coefficients, b_π and b_y , were loosened substantially – from Normal(2.10, 0.15) and Gamma(0.28, 0.10) to Gamma(1.00, 1.00) and Gamma(1.00, 0.70) respectively – so that the posterior is not mechanically dominated by a tight prior centered near the previous round’s own answer. A first attempt used Gamma(1.00, 1.00) (shape 1, i.e. Exponential(1)) for both; b_π ’s posterior moved away from the prior’s mode at zero without difficulty, but b_y ’s mode landed at exactly 0.0000 with a non-positive-definite Hessian there – a corner solution the shape-1 prior itself pre-loads, since an Exponential density is maximized at zero. b_y ’s prior was reshaped to Gamma(1.00, 0.70) (shape ≈ 2.04 , an interior mode near 0.51, no boundary pre-loading) while keeping the same diffuse mean, which resolved this. The fiscal-feedback parameter φ_b ’s prior mean was also lowered, from 0.10 to 0.05, keeping its prior standard deviation (0.05) unchanged.

1.1 Summary of stochastic elements

Table 1 consolidates every stochastic element in the model – the eight structural shocks discussed above and the seven measurement-error shocks attached to the observable equations – in one place, since the discussion above introduces each one individually at the point it enters the model rather than as a single reference list.

Symbol	Affects	Law of motion	Persistence
<i>Structural shocks</i>			
$\varepsilon_{a,t}$	Preference wedge a_t (eq. ??)	$a_t = a_{t-1}^{\rho_a} \exp(\varepsilon_{a,t})$	$\rho_a = 0.95$
$\varepsilon_{z,t}$	Productivity z_t (eq. prod)	$z_t = z_{t-1}^{\rho_z} z^{1-\rho_z} \exp(\varepsilon_{z,t})$	ρ_z estimated
$\varepsilon_{va,t}$	Shopping-time pref. $v_{a,t}$ (eq. hs)	$v_{a,t} = v_{a,t-1}^{\rho_{va}} v_a^{1-\rho_{va}} \exp(\varepsilon_{va,t})$	$\rho_{va} = 0$
$\varepsilon_{xa,t}$	Deposit technology $x_{a,t}$ (eq. d)	$x_{a,t} = x_{a,t-1}^{\rho_{xa}} x_a^{1-\rho_{xa}} \exp(\varepsilon_{xa,t})$	$\rho_{xa} = 0.95$
$\varepsilon_{g,t}$	Govt. spending g_t (resource constr.)	$g_t = \rho_g g_{t-1} + (1 - \rho_g)g + \varepsilon_{g,t}$	$\rho_g = 0$
ε_t^r	MP shock z_t^r (Taylor rule)	$z_t^r = b_r z_{t-1}^r + \varepsilon_t^r$	b_r estimated
ε_t^τ	Tax rule (eq. tau)	iid	–
ε_t^L	Labor-leisure FOC (COVID stand-in)	iid	–
<i>Measurement-error shocks (one per observable, all iid)</i>			
$\varepsilon_{y,t}^{me}$	y_t^{obs} (output)	–	–
$\varepsilon_{d,t}^{me}$	d_t^{obs} (deposits)	–	–
$\varepsilon_{\pi,t}^{me}$	π_t^{obs} (inflation)	–	–
$\varepsilon_{r,t}^{me}$	r_t^{obs} (loan rate)	–	–
$\varepsilon_{r^n,t}^{me}$	$r_t^{n,obs}$ (policy rate)	–	–
$\varepsilon_{\tau,t}^{me}$	τ_t^{obs} (tax rate)	–	–
$\varepsilon_{bT,t}^{me}$	bT_t^{obs} (debt/GDP)	–	–

Table 1: All fifteen stochastic elements of the estimated model. “Persistence” reports whether the process’s own AR(1) coefficient is fixed by calibration or freely estimated; iid processes (measurement errors and three of the structural shocks) have no persistence parameter. Standard deviations, priors, and posterior estimates for all fifteen are in Table 7.

Eight of the fifteen are structural, entering the model’s own equations and so shaping the dynamics of $y_t, \pi_t, r_t^\ell, \dots$ directly; seven are measurement error, entering only the observation equations that link model variables to the FRED data series (Table 2 below) and therefore affecting the likelihood but not the model’s own propagation mechanism. Of the eight structural shocks, persistence splits four ways: ε_t^τ and ε_t^L have no persistence parameter at all (direct

iid entry); $v_{a,t}$ and g_t are written with an AR(1) law of motion but have that persistence fixed at exactly zero (functionally iid); a_t and $x_{a,t}$ have persistence fixed at a nonzero value, $\rho_a = \rho_{xa} = 0.95$ (freeing them previously drove ρ_a to a near-unit-root value and produced a non-positive-definite Hessian, Section 2.2); and z_t and z_t^r have their persistence (ρ_z , b_r) freely estimated.

2 Estimation Methodology

2.1 Data

The model is estimated on seven quarterly U.S. observables spanning 2009Q1–2025Q4 (68 observations), all sourced from the Federal Reserve Bank of St. Louis FRED database. Series reported at a higher frequency (monthly or weekly) are averaged to quarterly. Every series is demeaned by its own full-sample mean prior to estimation (in logs, consistently with every measurement equation), matching the model’s measurement equations, which express each observable as a log-deviation from steady state plus measurement error. Table 2 lists the construction of each series; currency in circulation, used as an eighth observable in earlier versions of this model, is excluded here.

Observable	FRED series	Transformation
y^{obs} (output)	GDPC1 / OPHNFB	$\log(\text{GDPC1}/\text{OPHNFB})$, demeaned
d^{obs} (deposits)	DPSACBW027SBOG / OPHNFB	$\log(\text{Deposits}/\text{OPHNFB})$, demeaned
π^{obs} (inflation)	GDPDEF	$\log(\text{GDPDEF}_t/\text{GDPDEF}_{t-1})$, demeaned
r^{obs} (loan rate)	BAA	$\log(1 + \text{BAA}_q/400)$, demeaned
$r^{n,obs}$ (policy rate)	FEDFUNDS	$\log((1 + \text{FEDFUNDS}/100)^{0.25})$, demeaned
τ^{obs} (tax rate)	FGRECPT / GDP	$\log(\text{Tax receipts}/\text{Nominal GDP})$, demeaned
bT^{obs} (debt/GDP)	FYGFDPUN / GDP	$\log((\text{FYGFDPUN}/1000)/\text{GDP})$, demeaned

Table 2: Observable construction. OPHNFB is nonfarm business sector output per hour (labor productivity), used to detrend output and deposits. BAA (Moody’s Seasoned Baa Corporate Bond Yield, monthly) replaces the bank prime rate (MPRIME) used in earlier versions of this model. FYGFDPUN (Federal Debt Held by the Public, quarterly, millions of dollars) replaces GFDEBTN (gross/total public debt); it is divided by 1000 before forming the ratio because FYGFDPUN is reported in millions of dollars while GDP is reported in billions. q subscripts denote quarterly averages of monthly series.

Data-provenance notes. Two issues were found and fixed while rebuilding this observable. First, NGDPDQ – referenced as the GDP denominator in an earlier data-fetch script’s header comments and in this model file’s own prior header – is not a valid FRED series (`fredgraph.csv?id=NGDPDQ` returns an HTML error page, not data); the debt-to-GDP data actually used in earlier estimation rounds came from a separately pre-downloaded ratio file, not from that script. The nominal-GDP denominator is now GDP, the same series already used correctly elsewhere in this pipeline for τ^{obs} . Second, the measurement equation for bT_t^{obs} is a log-deviation, but the data-construction scripts used in earlier rounds demeaned the raw ratio in *levels* instead; the current build demeans in logs, matching the model.

2.2 Two-stage mode-finding

OccBin’s piecewise-linear likelihood is nonsmooth at ZLB regime switches, and gradient-based optimizers such as `csmnwel` tend to fail on it directly. Estimation therefore proceeds in two stages, repeated whenever the observation data *or* the calibration changes (mode-finding depends on the likelihood, which depends on both, even when the model equations themselves are unchanged):

1. **Stage 1 (linear Kalman filter).** With `options_.occbin.likelihood.status = false`, the posterior mode is located with `mode.compute=4` (`csmnwel`). An initial attempt that additionally freed the productivity persistence parameters ρ_a, ρ_{xa} (previously fixed at 0.95 in the baseline model) converged to $\rho_a \approx 0.9992$ – effectively a unit root – and produced a non-positive-definite Hessian at the mode; $\rho_a = \rho_{xa} = 0.95$ have been fixed (not estimated) ever since. Under the current calibration ($\bar{\tau} = 0.14, \chi = 0.5$), Stage 1 mode-finding again produced a positive-definite Hessian with no singularity warnings, acceptance ratios 0.223/0.230 on a short diagnostic run (informational only, since this stage still uses the linear, non-OccBin, approximation).
2. **Stage 2 (true OccBin likelihood).** With `options_.occbin.likelihood.status = true`, estimation is restarted at `mode.compute=0` using the Stage 1 mode as the starting point. A mode file computed under a different parameter vector *or* different data/calibration causes Dynare to throw “generated using another specification of the model” (in the parameter-vector case) or simply to start MCMC from a stale, no-longer-optimal point (in the data/calibration-change case, which Dynare’s compatibility check does not catch and must be tracked manually) – Stage 1’s mode, rebuilt under the identical parameter vector and current data/calibration, avoids both.

2.3 MCMC jumping covariance and jump-scale tuning

An initial production run under the previous calibration ($\bar{\tau} = 0.20, \chi = 1.0, \text{mh_jscale}=0.02, \text{mcmc_jumping_covariance}=\text{prior_variance}$) passed the 0.05-acceptance-ratio floor (0.271/0.242) but produced posterior densities for three of the four policy parameters (b_π, b_y, φ_b) that visibly appeared bimodal when the two chains’ draws were pooled. Checking the two chains separately showed this was chain disagreement, not genuine multimodality (chain means 1.86 and 1.67 pooled standard deviations apart for b_π, b_y respectively, each chain individually unimodal and tight) – consistent with `prior.variance`’s diagonal proposal covariance ignoring the b_π/b_y correlation inherent to Taylor-rule estimation.

Switching `mcmc\_jumping\_covariance` to `hessian` (using the Stage-1 mode’s Hessian, which captures this correlation) fixed the chain-disagreement problem in the subsequent round (under $\bar{\tau} = 0.20$, FYGFDPUN-based data), but required extensive `mh\_jscale` re-tuning from scratch, including one attempt (`mh\_jscale=1.0`) that produced 6,861 near-singular-matrix warnings and took over an hour without finishing 3,000 draws, and repeated cases where short (3,000-draw) diagnostics overstated what a full 50,000-draw production run would sustain. Table 3 reports the full history across all rounds; for the current calibration ($\bar{\tau} = 0.14, \chi = 0.5$), the `mh\_jscale=0.10` setting that had worked previously was tried directly as an informed first guess and succeeded on the first production attempt (0.1775/0.1534), the first round not requiring a kill-and-retune cycle.

Jumping covariance	mh_jscale	Chain 1	Chain 2	Outcome
$\bar{\tau} = 0.20, \chi = 1.0$; <i>debt observable: GFDEBTN, level-demeaned</i>				
prior_variance	0.05	0.0433	0.0710	Chain 1 below 0.05 – killed
prior_variance	0.03 (diag., 3k)	0.203	0.057	Both above 0.05 – promoted
prior_variance	0.03 (prod., 50k)	0.157	0.0477	Chain 2 below 0.05 – killed
prior_variance	0.02 (diag., 3k)	0.306	0.293	Both healthy – promoted
prior_variance	0.02 (prod., 50k)	0.271	0.242	Accepted; bimodal pooled densities
$\bar{\tau} = 0.20, \chi = 1.0$; <i>debt observable: FYGFDPUN/GDP, log-demeaned</i>				
hessian	1.0 (diag., 3k)	0.0073	0.0040	Both far below 0.05; 6,861 singular warnings, >1hr – killed
hessian	0.2 (diag., 3k)	0.100	0.104	Both healthy on diagnostic – promoted
hessian	0.2 (prod., 50k)	0.0449	0.0582	Chain 1 below 0.05; took 15h03m – killed
hessian	0.12 (diag., 3k)	0.122	0.129	Both healthy – extra margin sought
hessian	0.10 (diag., 3k)	0.236	0.265	Both in textbook 0.2–0.4 range – promoted
hessian	0.10 (prod., 50k)	0.214	0.195	Accepted
$\bar{\tau} = 0.14, \chi = 0.5$; <i>debt observable: FYGFDPUN/GDP, log-demeaned</i>				
hessian	0.10 (diag., 3k)	0.405	0.259	Both healthy, good margin – promoted
hessian	0.10 (prod., 50k)	0.1775	0.1534	Accepted, but Geweke uneven (§5)
posterior-cov	0.5 (diag., 3k)	0.2047	0.2017	Textbook range, symmetric – promoted
posterior-cov	0.5 (prod., 50k)	0.0805	0.0387	Chain 2 below 0.05 – killed
posterior-cov	0.25 (diag., 3k)	0.543	0.441	Above textbook range, extra margin sought
posterior-cov	0.25 (prod., 50k)	0.231	0.114	Accepted

Table 3: Full mh_jscale/jumping-covariance/calibration tuning history. `posterior-cov` denotes a custom jumping covariance built from a preliminary round’s own pooled posterior draws (see Section 2.4). The final production run (`posterior-cov`, `mh_jscale`=0.25, $\bar{\tau} = 0.14$, $\chi = 0.5$) completed in 7h51m.

2.4 Jumping covariance from pooled posterior draws

The `hessian`-covariance production run under the current calibration passed the acceptance-ratio floor (0.1775/0.1534) but showed renewed chain disagreement: Geweke’s 15% taper test passed on only 1 of 8 structural parameters for chain 2 (versus 6 of 8 for chain 1), and the chain-mean gaps for b_y and a_i were 0.95 and 0.96 pooled standard deviations, respectively (Section 5 reports the full comparison across all three jumping-covariance choices). Rather than switch back to `prior_variance` – which would reintroduce the correlation-blindness mechanism that caused the *original* chain-disagreement problem – a custom jumping covariance was built directly from that run’s own pooled, post-burn-in posterior draws (`build_posterior_jumping_covariance.m`), under the reasoning that the empirical posterior covariance reflects the model’s actual local geometry better than a Hessian evaluated only at the mode of a preliminary (linear-Kalman-filter) approximation.

One implementation subtlety, verified directly against Dynare 7.1’s source (`set_mcmc_jumping_covariance.m`) despite the required variable name `jumping_covariance`, a user-supplied matrix loaded via this mechanism is treated as a Hessian-like precision object and is *inverted* internally before use as the actual proposal covariance. To make the sampler propose from the empirical posterior covariance $\hat{\Sigma}$, the file must therefore store $\hat{\Sigma}^{-1}$, not $\hat{\Sigma}$ itself.

The starting point (`mode_file`) was left unchanged – still the Stage-1 linear-Kalman-filter mode, a valid parameter vector under the current calibration – so only the proposal *shape* changed. Using the classic Roberts-Gelman-Gilks scaling heuristic $2.4/\sqrt{23} \approx 0.5$ as a starting

`mh_jscale` (appropriate here since this covariance reflects actual posterior spread rather than local mode curvature, unlike the Hessian, which needed a very different `mh_jscale=0.10`), a 3,000-draw diagnostic gave an excellent, symmetric 0.2047/0.2017 acceptance ratio. One parameter, a_i , showed a large chain-mean gap in that diagnostic (1.83 pooled standard deviations, both chains far from the true posterior mean of ≈ 0.18); given the diagnostic kept only 1,500 draws per chain after burn-in from a mode-file starting point found under the *linear* approximation, this was interpreted as insufficient travel time from a mismatched starting point rather than genuine disagreement, and the run was promoted directly to a full production attempt. That interpretation was confirmed: in the full run (25,000 kept draws per chain), a_i 's chain-mean gap fell to 0.02 pooled standard deviations (Section 5 reports the complete comparison for all four policy parameters). The first full production attempt at `mh_jscale=0.5` failed the acceptance floor on chain 2 (0.0805/0.0387) despite the healthy diagnostic – consistent with the diagnostic-overstates-production pattern seen throughout this estimation exercise – and a second attempt at `mh_jscale=0.25` (after an intermediate 3,000-draw diagnostic at that setting, which gave 0.543/0.441, above the textbook range but promoted directly given the steepness of the degradation just observed) was accepted at 0.231/0.114.

The final production run used `mh_jscale=0.25`, `mcmc_jumping_covariance` pointing at the custom posterior-covariance file, `mh_replic=50,000`, `mh_nblocks=2`, `mode_compute=0`, completing in 7 hours 51 minutes. The Modified-Harmonic-Mean log marginal data density is 1151.14.

2.5 Smoother limitation

OccBin's post-estimation nonlinear smoother (both the iterative and conditional variants) again did not reach a fixed point for the smoothed ZLB regime path, so no OccBin-consistent smoothed shock or state series are available (a linear-smoother fallback completed but is not ZLB-consistent and is not reported here). This does not affect the posterior parameter estimates below, which are informed by OccBin's *filter* (the forward likelihood pass used throughout MCMC, which did enforce the ZLB across the full sample) rather than the smoother.

Which quarters does the estimation flag as ZLB-binding? The saved results do store OccBin's regime history (`oo_.occbin.smoother.regime_history` and `realtime_regime_history`), and Figure 1 compares it with a model-independent check: quarters in which the linear smoother's notional rate $r_t^{n,notional}$ falls below the floor of 1. The two do not agree, and the OccBin path should not be taken at face value. The OccBin history flags 2009Q1–2019Q4 (44 quarters) and 2020Q2–2022Q1 (8 quarters) as binding, and its real-time and smoothed histories are identical. This is Dynare's fallback: `warning_flag = 322` means the smoother did not reach a fixed point and the reported smoothed regimes are replaced by the real-time (forward filter) path. Re-running the smoother at the posterior mean with more iterations, a larger solver budget, and the data-implied ZLB dates as the starting regime guess left this outcome unchanged (tightening the solver made it fail outright), so the saved path is the filter's own. Binding through 2019 conflicts with the data (the policy rate was well above zero from late 2016), and the OccBin smoother's own notional-rate path is erratic (it swings above 1.02 in 2021 while flagged as binding), consistent with the convergence warnings the smoother raised (`warning_flag = 322`) and with the fixed-point failure described above. The linear check instead gives 2009Q3–2016Q3 and 2020Q3–2022Q1, much closer to the 2009Q1–2015Q4 and 2020Q1–2021Q4 episodes cited in Section 1; the two methods differ in 15 quarters. Note that the linear check is a fragile threshold test: in 2009–2016 the notional rate sits within about 10^{-3} of the floor. The episode dates cited

in Section 1 come from the observed policy-rate series rather than from the estimated regime path, and this estimation cannot pin the binding quarters down more sharply.

The floor itself is not estimated: `rn.lb = 1` is calibrated, and the `occbin_constraints` block switches regimes when $\log r_t^{n,notional} < -10^{-6}$. One possible source of the misclassification is a mismatch between that floor and the demeaned observable, since a model rate of 1 corresponds to $r_t^{n,obs} = -\log(1.002214) = -0.00221$. Checking directly, $r_t^{n,obs}$ averages -0.00293 over 2009–2015 and -0.00304 over 2020Q2–2021Q4, implying model rates of 0.99928 and 0.99917: about 0.0007 below the floor, less than the estimated measurement-error standard deviation of $r^{n,obs}$ (about 0.003). The demeaning is therefore roughly consistent with the floor and does not explain the misclassification. In 2017–2019, $r_t^{n,obs}$ averages $+0.0009$ (implied rate 1.0031), well above the floor, so the OccBin binding flag there is not driven by the observed policy rate. Because the real-time (forward filter) history flags the same quarters, and the likelihood is built from that pass, the flag cannot be attributed to the smoother’s non-convergence alone.

Decomposing the rule at the posterior mean identifies the cause. Binding requires $b_\pi \Delta \log \pi_t + b_y \Delta \log y_t + z_t^r < -0.0022$, the gap between the steady-state policy rate (1.0022, 0.89% annualized) and the floor. Averaged over 2017–2019 the three terms are -0.0069 (inflation), $+0.0022$ (output), and -0.0002 (monetary shock), a total of -0.0048 : the inflation term alone pushes the notional rate below the floor, and with $b_\pi = 3.47$ an inflation shortfall of only 0.25 percentage points (annualized) below steady state suffices. The steady state is pinned to the sample mean, which includes the 2021–23 inflation surge, so 2017–2019 inflation (near the 2% target) appears low. The filter is therefore behaving as the model implies: a strong, non-inertial rule with a small cushion above the floor calls for negative rates in those years, and the observed positive rates are absorbed by measurement error in $r^{n,obs}$ (about one standard deviation there).

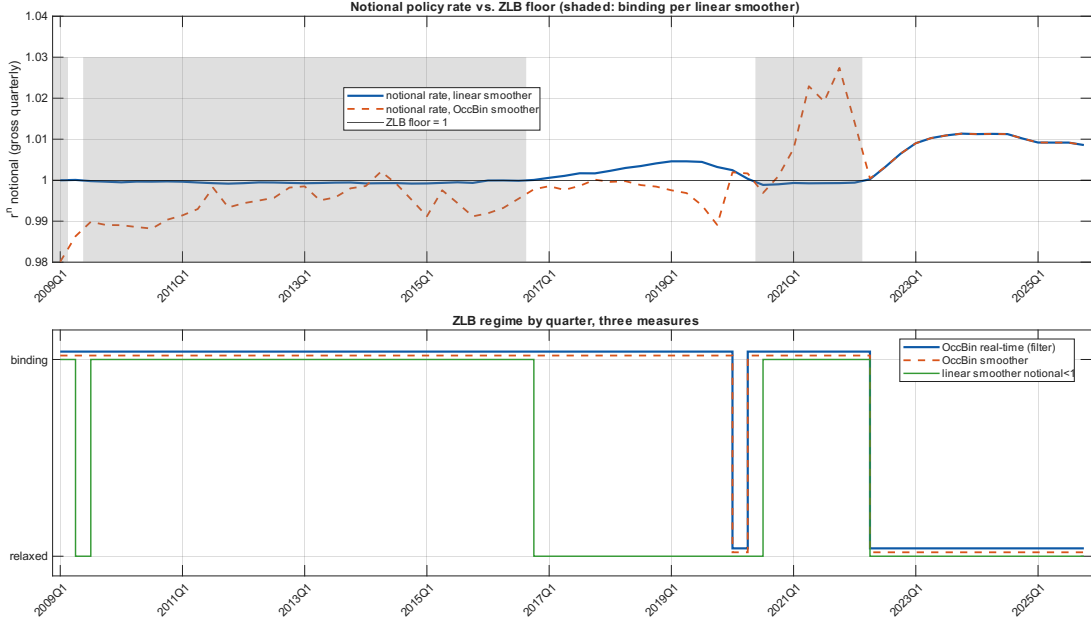


Figure 1: Top: notional policy rate from the linear and OccBin smoothers against the ZLB floor of 1 (shaded: quarters in which the linear smoother’s notional rate is below 1). Bottom: ZLB regime by quarter under the three measures.

Is the 2017–2019 binding self-reinforcing? Reserves n_t and currency m_t are not observables (Table 2), and the model-implied values in 2017–2019 are close to steady state (n_t averages 9.45 vs. 9.38, m_t 9.51 vs. 9.46); neither enters the policy rule directly. To bound any indirect channel through them, we treated binding as an additional policy-rate shock in the linear model at the posterior mean, of size $-\log r_t^{n,notional}$ in each binding quarter (average 0.0026 over 2016Q4–2019Q4, peak 0.011), and measured its total feedback on the other variables, averaged over 2017–2019. The effects on n_t and m_t are negligible (+0.16% and +0.08% of steady state) and on y_t essentially zero, but inflation is lower by 0.00099 per quarter (0.4 percentage points annualized) and the loan rate by 0.0008: holding the rate at the floor when the rule calls for less keeps policy tighter than it would otherwise be. With $b_\pi = 3.47$ that lower inflation reduces the notional rate by a further ≈ 0.0034 , against an average 2017–2019 shortfall of -0.0048 relative to the binding threshold of -0.0022 . Without the feedback the shortfall would be about -0.0014 , i.e. not binding, so the spell may be partly self-sustaining: binding depresses inflation enough to justify itself. The carry-over of the genuine 2009–2016 and 2020–22 spells into 2017–2019 is essentially zero. This is a first-order, unanticipated-shock approximation of what OccBin does with anticipated spells, so it suggests the possibility rather than proving it. A direct test was inconclusive: evaluating the OccBin filter and smoother at the posterior mean from four different initial regime guesses (default relaxed; binding through 2016Q3 and 2020Q2–2022Q2 with 2016Q4 onward relaxed; one long spell through 2019Q4; all relaxed) gives an identical log posterior (1266.559), `warning_flag`, and regime path in every case, so the filter either reaches the same fixed point from any start or ignores the guess, and these runs cannot distinguish the two.

Does a ZLB equilibrium exist at all? The regime-search failures above raise a more basic question: does a well-defined equilibrium in which the floor genuinely binds exist for this model at these parameters, independent of whether OccBin’s switching search can find it? This is tested directly, rather than inferred from the switching search’s behavior, using a companion file (`mmt8bOccBin3_covid.bT_baa.bindonly.mod`, built the same way as the existing `_bkonly.mod` determinacy companion) that imposes $r_t^n = \underline{r}^n$ *permanently* as the only monetary policy equation – no `occbin.constraints` block, no switching – and asks whether that system, on its own terms, has a steady state and whether that steady state is determinate. At the posterior mean, a steady state does exist and is economically unremarkable: $y = 1.020$, $c = 0.820$, $h = 0.943$, $\pi = 1.00653$ (2.63% annualized), $r^\ell = 1.010$, and $r^{n,notional} = 1.00181$ (the notional rate’s own equation is left active, unassigned to r_t^n , purely for reference). But Blanchard-Kahn’s order condition fails at this steady state: only 3 eigenvalues lie outside the unit circle against 4 forward-looking variables, so the permanently-bound system is *locally indeterminate*. This is the standard ZLB-indeterminacy mechanism – pinning r_t^n at a constant removes the Taylor-rule feedback that normally rules out self-fulfilling inflation paths – and it matches Ascari, Mavroeidis, and McClung’s (2023) result below, not an anomaly specific to this model. A further inconsistency: $r^{n,notional} = 1.00181$ at this fixed point is *above* the floor, so the same equilibrium fails OccBin’s own bind/relax test ($\log r^{n,notional} < 0$) – the policy rule itself would not choose to bind here, so this permanently-bound fixed point is not self-justifying in the sense the switching search requires. Taken together: a bind-regime equilibrium exists, so non-existence alone does not explain the search failures, but it is indeterminate and not obviously reconcilable with the regime-consistency condition OccBin’s switching search imposes – a concrete, model-specific instance of exactly the kind of difficulty the published literature below documents can arise near the ZLB.

Attempts to add interest-rate smoothing. Since a non-inertial rule is the source of the 2017–2019 flag, we tried adding smoothing that operates only away from the ZLB (“Version A”):

$$r_t^{n,notional} = (r_{t-1}^n)^{\rho_r} \left[(r_t^\ell)^{a_i} \left(\frac{\pi_t}{\pi} \right)^{b_\pi} \left(\frac{y_t}{y} \right)^{b_y} e^{z_t^r} \right]^{1-\rho_r},$$

with the lag of the *actual* rate entering only the relaxed-regime rule (the binding regime, $r_t^n = 1$, ignores it). Model files are `mmt8b0ccBin3_covid_bT_baa_smooth*.mod`; the original files are unchanged. The results were negative.

Stage 1 (linear filter). Smoothing is strongly favored: with a Beta(0.75, 0.10) prior the mode is $\rho_r = 0.955$ (s.d. 0.014) with $b_\pi = 7.86$, and the Laplace log data density rises from 1113.7 to 1174.9 (a gain of about 61 for one parameter); a Uniform(0.01, 0.99) prior gives $\rho_r = 0.967$ (s.d. 0.011), $b_\pi = 9.00$, log density 1177.0, so the high persistence is not a prior artifact. The other parameters barely move. a_i is not identified under the linear filter (it returned its prior mean, 0.49, and shifts only the steady-state rate), so it was fixed at 0.181 for these runs. With $a_i = 0.181$ the notional rate is below 1 in 2009Q3–2016Q1 and 2020Q3–2022Q1 and never in 2017–2019.

Stage 2 (OccBin likelihood). The OccBin filter, however, returns an infinite likelihood at the Stage-1 mode (it cannot find a consistent regime path). Evaluating on a grid around the mode, the likelihood was finite at only 47% of points ($\rho_r \in [0.93, 0.97]$, $b_\pi \in [5, 11]$) and at no point for $\rho_r = 0.97$; a `csmine1` search from a finite start stalled with a singular Hessian. Table 4 profiles the log posterior over ρ_r , maximizing over b_π , b_y , σ_r , and σ_{me,r^n} (Nelder–Mead, other parameters at their Stage-1 values). No finite point was found for $\rho_r \leq 0.5$, and the best value, 1260.0 at $\rho_r = 0.95$, is below the 1266.2 of the original (no-smoothing) OccBin mode, although only four of 23 parameters were re-optimized.

ρ_r	Best log posterior	b_π	$(1 - \rho_r) b_\pi$
≤ 0.5	no finite point	–	–
0.80	1256.8	2.82	0.56
0.90	1256.6	4.71	0.47
0.95	1260.0	6.95	0.35

Table 4: Profile of the OccBin log posterior over ρ_r (see text). The original no-smoothing model attains 1266.2 at its own mode.

At the best point ($\rho_r = 0.95$), the real-time regime path removes the spurious 2017–2019 binding (none of those quarters is flagged), but it flags only 21 of the 41 quarters in which the observed policy rate was at the floor (2009Q1–2016Q4 and 2020Q2–2022Q2): the 2009–2015 spell is broken into scattered pieces and only 2 of the 9 quarters in 2020Q2–2022Q2 are flagged (2021 is entirely relaxed), with the near-zero observed rate absorbed by $r^{n,obs}$ measurement error. Smoothing therefore trades one misclassification for another and does not improve the OccBin fit; the gain seen under the linear filter appears to reflect smoothing standing in for the ZLB when the ZLB is ignored. We retain the original specification and treat the over-extended 2017–2019 binding as a documented limitation. Its effect on the reported posterior, in particular on b_π and a_i , has not been quantified.

Attempts to re-anchor the policy rule to a fixed 2% inflation target. Section 2.5 traces the 2017–2019 misclassification to the rule’s inflation term being measured against the

model’s own endogenous steady state, $\log r_t^{n,notional}$ falling below the floor whenever inflation dips modestly below that (elevated, deficit- financed) benchmark rather than below a conventional target. A natural fix is to reference a fixed 2% annual target instead: the calibration block already defines $\pi^{ss} = 1.02^{0.25} = 1.004963$ for exactly this purpose (Table 6), but it was, in the specification examined above, used only in the tax-rule/debt-issuance block, *not* in the policy rule itself, which used `STEADY_STATE(pig)`. Replacing `STEADY_STATE(pig)` with π^{ss} in

$$r_t^{n,notional} = (r^{\ell,ss})^{a_i} \left(\frac{\pi_t}{\pi^{ss}} \right)^{b_\pi} \left(\frac{y_t}{y^{ss}} \right)^{b_y} e^{z_t^r}$$

and re-solving the steady state at the current posterior mean (all other parameters unchanged) produces a steady state with $\pi_{ss} = 0.99899$ quarterly (-0.41% annualized – mild deflation, not 2% inflation) and $r_{ss}^{n,notional} = 0.99223$: *below* the ZLB floor of 1 in the model’s own deterministic steady state. Blanchard-Kahn order and rank conditions are still satisfied (4 forward-looking eigenvalues), so the linearized system is well-posed, but a steady state that already violates the floor is not economically usable for an occasionally-binding-constraint model, whose premise is that the floor is an occasional, shock-driven excursion away from a steady state comfortably above it. A sweep of $a_i \in \{0.184, 0.30, 0.49, 0.65, 0.80, 0.95, 0.99\}$ (holding all other parameters fixed) left π_{ss} , r_{ss}^{ℓ} , and $r_{ss}^{n,notional}$ completely unchanged to five decimal places: a_i enters the rule only as the exponent on $(r^{\ell,ss})^{a_i}$, and since $r^{\ell,ss} \approx 1.00229$ in this steady state, raising a number that close to 1 to any power between 0 and 1 moves the result by at most ≈ 0.02 percentage points – roughly 1/40 of the ≈ 0.8 percentage point gap that would need to be closed. The mechanism is not a fixable calibration choice but a genuine tension between the model’s seigniorage need (financing the calibrated deficit, which is what pushes the *endogenous* steady state to 3.63% annualized in the reported specification) and a rule pinned to a conventional 2% target evaluated at policy coefficients ($b_\pi = 3.36$, $a_i = 0.184$) estimated under the old rule. Resolving it would require re-estimating b_π, b_y jointly with the new rule specification, or revising the fiscal calibration $(\bar{\tau}, g)$ that drives the seigniorage need, with no guarantee either produces a steady state clearing the floor by a sensible margin. This was not pursued further.

Is this a known difficulty? Yes. Boehl (2022), benchmarking a faster alternative algorithm against OccBin on a medium-scale New Keynesian model with a single occasionally-binding interest-rate lower bound – the same class of constraint as here – reports that OccBin’s Newton-like solver (Guerrieri and Iacoviello, 2015) “can not draw on the nice convergence properties of the regular real-valued Newton method,” which “renders their method impractical once OBC problems are not well behaved,” and quantifies actual failures in a benchmark run: two of the sampled parameter draws returned no solution at all for OccBin, and the paper notes generally that “this causes a very small number of samples to not converge.” This is a published, quantified precedent for OccBin’s regime-consistency search failing outright, in the same single-ZLB-bound setting used here, not merely a difficulty specific to this model. A related but distinct concern is theoretical rather than algorithmic: Ascari, Mavroeidis, and McClung (2023) show that standard rational-expectations models with an occasionally-binding ZLB constraint can, at some parameter or shock configurations, have no solution at all or multiple solutions – raising the possibility that some of the points where OccBin’s search fails have no consistent regime path to find in the first place, not merely one the algorithm cannot locate.

Implications for the reported posterior. The difficulties documented in this subsection – the 2017–2019 misclassification, its partial and only partially successful remediation attempts,

and the OccBin likelihood’s general fragility near this boundary (Sections 2.5–8.4) – are specifically failures of OccBin’s piecewise-linear regime-consistency search, not of the underlying Bayesian estimation. Every retained MCMC draw, by construction, has a successfully evaluated OccBin likelihood (that is how it was accepted), so the posterior in Table 7 reflects genuine exploration under the true (ZLB-consistent) likelihood, not a fallback linear approximation. What does *not* carry over automatically is evaluating the model at a single representative point such as the posterior mean: that point is a purely arithmetic average of the retained draws, never itself evaluated during estimation, and it can (and in exercises reported elsewhere did) fall where OccBin’s regime search fails outright even though every draw contributing to the average was individually evaluable. This does not affect the parameter posterior itself, nor any exercise conducted on the model’s plain linear (non-ZLB, “RELAX”-regime) solution – verified directly by re-solving that linear system at a posterior mean where the OccBin filter failed, which converged cleanly with Blanchard-Kahn conditions satisfied. The determinacy/E-stability analysis (Section 6) and the counterfactual and decomposition exercises (Section 8.1) all use this linear solution and are unaffected; only OccBin-specific exercises evaluated at the posterior mean (rather than at an individual draw, such as the posterior mode) are at risk.

An alternative: smoothing the floor instead of switching regimes. A structurally different way to avoid OccBin’s regime-consistency search altogether is to replace the piecewise floor with a smooth approximation of it, then estimate the resulting (kink-free) nonlinear model with a standard higher-order perturbation plus particle-filter likelihood instead of OccBin’s. This is a documented approach: Andreasen and Kronborg (2022, *Quantitative Economics* 13, 1171–1202) solve a New Keynesian model with the ZLB imposed via “a smooth nonlinear modification of the standard Taylor rule,” paired with their “extended perturbation” method and a particle filter; Richter and Throckmorton (2015, *The B.E. Journal of Macroeconomics* 15(1), 157–182) discuss smoothing the floor for global-solution numerical convergence. Both papers’ exact functional forms could not be extracted here – both PDFs obtained were scanned or access-protected – so the specific logistic form used below is our own construction, not a transcription of either paper’s calibrated smoothing parameter.

A companion file, `mmt8bOccBin3.covid.bT.baa.softzlb.mod`, replaces the `occbin.constraints` block with a single equation:

$$r_t^n = \underline{r}^n + \frac{r_t^{n,notional} - \underline{r}^n}{1 + \exp(-\kappa(r_t^{n,notional} - \underline{r}^n))},$$

a logistic soft-max of $(r_t^{n,notional}, \underline{r}^n)$ that converges to $\max(r_t^{n,notional}, \underline{r}^n)$ – OccBin’s exact floor – as $\kappa \rightarrow \infty$. With $\kappa = 2000$ the transition width (roughly $10/\kappa$ in gross-quarterly-rate units, ≈ 0.005) is not negligible relative to the steady-state gap above the floor ($r_{ss}^n - \underline{r}^n \approx 0.0022$ at the posterior mean), producing a small steady-state “shaving” artifact ($r_{ss}^n = 1.00218$ against $r_{ss}^{n,notional} = 1.00221$); κ remains an uncalibrated free choice to sensitivity-check, not a settled value. At these parameters the smoothed model’s steady state is well-defined and the Blanchard-Kahn order and rank conditions are verified (4 eigenvalues outside the unit circle for 4 forward-looking variables) – a cleaner local property than the permanently-bound piecewise system above, though the two are different questions (a linearization around the smoothed model’s own near-floor steady state, versus the piecewise system pinned exactly at the floor).

Particle-filter mechanics: two implementation findings. Getting this model estimated with Dynare’s particle filter (`options_.particle.status`, dispatching to `non_linear_dsge_likelihood.m`) surfaced two points worth recording for anyone extending this line of work. First, Dynare’s particle-filter initial checks (`initial_estimation_checks.m:116--120`) require measurement error registered in the observation-noise matrix M_H specifically; the ε_t^{me} terms in this project’s measurement equations are ordinary structural shocks loading through Σ_e , invisible to that check even though they are the same additive iid noise econometrically. The diagnostic file therefore declares measurement error the standard Dynare way (`var y_obs; stderr ...`; in the `shocks` block) rather than as explicit `eps_*_me` shocks – a mechanical reformulation, not a modeling change.

Second, and more consequential: Dynare’s Monte-Carlo particle filters (`sequential_importance_particle_filter.m:72`, and identically in the auxiliary and conditional particle filters) call `set_dynare_seed_local_options([],false,'default')` at the start of every likelihood evaluation, which unconditionally resets MATLAB’s random-number generator to a fixed seed – confirmed directly by evaluating the likelihood at identical parameters under three different `set_dynare_seed()` calls and obtaining bit-identical results to 12 decimal places. This is a deliberate common-random-numbers design: it makes the particle-filter likelihood a smooth, reproducible function of the parameters, which helps an optimizer (though Dynare’s own mode-search warning, quoted below, is right that a gradient-based one still struggles with residual resampling-induced kinks) but is the wrong property for Metropolis-Hastings estimation. Valid particle-MCMC (Andrieu and Roberts, 2009, *Annals of Statistics* 37(2), 697–725) requires the likelihood estimator to be unbiased *and independently redrawn at every iteration*; with the RNG frozen to the same seed on every call, an MH chain built directly on this filter would repeatedly evaluate the same frozen particle realization rather than integrating out fresh approximation noise, and would not carry the usual guarantee of targeting the true posterior. Consequently, only mode-finding (not MCMC) is attempted below.

Mode comparison. `mode_compute=4` (gradient-based) immediately triggers Dynare’s own interactive warning that “particle filtering introduces discontinuities in the objective function w.r.t the parameters” and recommends a derivative-free method; `mode_compute=8` (Dynare’s own Nelder-Mead simplex) is used instead. Starting from OccBin’s own posterior mode – not the posterior mean of Table 7, but the mode found by this production round’s own mode-search step ($-\log \text{posterior} = 1291.297$) – the smooth-ZLB/particle-filter mode search (`mmt8bOccBin3_covid_bT_baa_softzlb_mode.mod`) converges, at 5,000 particles, to:

Parameter	OccBin mode	Smooth-ZLB mode (5,000 part.)	Diff
v^a	2.7466	2.8497	+3.8%
η	1.2596	1.2802	+1.6%
b_π	3.2795	3.2547	-0.8%
b_y	0.0721	0.0727	+0.8%
a_i	0.1809	0.1843	+1.9%
b_r	0.2554	0.2618	+2.5%
ρ_z	0.5026	0.5088	+1.2%
φ_b	0.0938	0.0942	+0.4%
σ_a	0.0195	0.0177	-9.3%
σ_z	0.0241	0.0244	+1.4%
σ_{va}	0.0463	0.0490	+5.7%
σ_{xa}	0.0421	0.0429	+1.9%
σ_g	0.0295	0.0301	+2.3%
σ_r	0.0180	0.0183	+2.0%
σ_τ	0.0315	0.0320	+1.7%
σ_L	0.0636	0.0647	+1.7%

Every structural parameter – including b_π , b_y , and a_i , which jointly govern how aggressively the rule responds and how close to the floor it gets – lands within about 1–4% of OccBin’s mode. Most shock standard deviations agree similarly closely; the two exceptions are σ_a (−9.3%) and σ_{va} (+5.7%), moving in opposite directions, plausibly because all seven measurement-error standard deviations are calibrated (fixed at OccBin’s own mode values) rather than re-estimated here, so any residual adjustment the smoothed model wants in fitting y_{obs} and d_{obs} has nowhere to go except into the structural shocks loading on the same observables. As a stability check, the same search was first run at 1,000 particles (for speed) and then warm-started at 5,000: every parameter moved by less than 1.6% between the two, and the achieved – log posterior rose from 1093.156 to 1182.373 – moving closer to OccBin’s 1291.297 as expected from a less biased (more particles) likelihood approximation, though this level is not otherwise comparable to OccBin’s: different filter, different approximation order, and measurement error entering through M_H rather than as structural shocks all change what is being evaluated.

This is a real cross-validation, not a replacement for the estimation reported in this paper: two structurally different likelihood-evaluation methods – OccBin’s piecewise-linear regime-switching Kalman filter/smoother and a smooth logistic floor with second-order perturbation and a particle filter – land in essentially the same corner of parameter space, at a point (the posterior mean) where the OccBin-specific machinery itself fails to evaluate. Given the seeding limitation above, this is offered as a mode comparison only; a genuine MCMC exploration under the smooth specification would first require patching Dynare’s particle filters to draw fresh randomness per iteration.

Sensitivity to κ . $\kappa = 2000$ above is an uncalibrated free choice, so it is checked two ways. First, a steady-state-only sweep (`mmt8b0ccbin3_covid_bT_baa_softzlb_kappasweep.mod`, no estimation) at OccBin’s posterior mode:

κ	r_{ss}^n	$r_{ss}^{n,notional}$	Gap	BK order+rank
1,000	1.001944	1.002167	0.000223	verified
2,000	1.002184	1.002210	0.000026	verified
5,000	1.002216	1.002216	0.00000003	verified
10,000	1.002216	1.002216	0.0	verified
20,000	1.002216	1.002216	0.0	verified

The shaving artifact shrinks fast and is essentially gone by $\kappa = 5,000$; $\kappa = 1,000$ carries about 8.5 times $\kappa = 2,000$'s artifact (still small in absolute terms, $\approx 0.02\text{pp}$ annualized). Determinacy holds at every value tested.

Second, a full mode search at $\kappa = 1,000$ (`mmt8b0ccBin3_covid_bT_baa_softzlb_mode_k1000.mod`, 1,000 particles, cold-started from OccBin's mode – identical protocol to the original $\kappa = 2,000$, 1,000-particle run) converges, in 573 iterations, to $-\log$ posterior = 1097.084 (against 1093.156 for $\kappa = 2,000$ under the same protocol – not comparable to OccBin's 1291.297, but directly comparable to each other since particle count, starting point, and every other choice are held fixed):

Parameter	OccBin mode	$\kappa = 1,000$ mode	$\kappa = 2,000$ mode
v^a	2.7466	2.7248 (-0.8%)	2.8445 (+3.6%)
η	1.2596	1.2388 (-1.6%)	1.2703 (+0.8%)
b_π	3.2795	3.2029 (-2.3%)	3.3034 (+0.7%)
b_y	0.0721	0.0743 (+3.1%)	0.0729 (+1.1%)
a_i	0.1809	0.1840 (+1.7%)	0.1824 (+0.8%)
b_r	0.2554	0.2559 (+0.2%)	0.2577 (+0.9%)
ρ_z	0.5026	0.5105 (+1.6%)	0.5078 (+1.0%)
φ_b	0.0938	0.0926 (-1.3%)	0.0947 (+1.0%)
σ_a	0.0195	0.0196 (+0.5%)	0.0177 (-9.2%)
σ_z	0.0241	0.0243 (+0.8%)	0.0243 (+0.8%)
σ_{va}	0.0463	0.0462 (-0.2%)	0.0490 (+5.8%)
σ_{xa}	0.0421	0.0432 (+2.6%)	0.0429 (+1.9%)
σ_g	0.0295	0.0300 (+1.7%)	0.0301 (+2.3%)
σ_r	0.0180	0.0183 (+1.7%)	0.0183 (+2.0%)
σ_τ	0.0315	0.0307 (-2.5%)	0.0320 (+1.7%)
σ_L	0.0636	0.0629 (-1.1%)	0.0647 (+1.7%)

κ is not a free lunch, but it also does not move the qualitative conclusion, and not always in the direction one might expect. Most structural parameters are comparably close to OccBin's mode under either κ (within roughly 1–3%). The two shock standard deviations that were $\kappa = 2,000$'s largest outliers, σ_a (−9.2%) and σ_{va} (+5.8%), are essentially resolved at $\kappa = 1,000$ (+0.5% and −0.2%), and the achieved $-\log$ posterior is *higher* at $\kappa = 1,000$ despite it being the cruder approximation of the true kink and carrying the larger steady-state artifact above. A plausible explanation is that the smoother transition at $\kappa = 1,000$ gives the particle filter an easier, less kinked surface to track through the actual ZLB episodes in the data – where most of the likelihood's information is – more than offsetting the larger (but still small) steady-state distortion; this is offered as a suggestive pattern from one comparison, not a settled explanation.

The practical conclusion is that sharpening κ is not obviously an improvement, and the smooth-ZLB cross-validation above is not an artifact of the particular κ chosen.

Could the smooth-ZLB model be pushed from a mode comparison to a full posterior? The seeding limitation identified above was pursued to see how far it could actually be taken, rather than left as a named caveat. Three steps were tried, in increasing order of ambition.

Patching the seed. A project-local shadow copy of `sequential_importance_particle_filter.m` (added to the MATLAB path ahead of the Dynare installation, which is left untouched) replaces the fixed-seed reset with a fresh, monotonically-incrementing seed per call. This was verified to work directly: the same bit-identical-to-12- decimals test that first exposed the problem now returns six genuinely different values across six calls at identical parameters.

Measuring the resulting noise. With independent draws restored, repeated likelihood evaluations at the posterior mean give the estimator’s true sampling variance: 1121 at 5,000 particles, falling to 104 at 50,000, consistent with the particle filter’s expected $1/N$ scaling (variance $\times N$ is roughly constant across that range, itself confirming the patch produces statistically sensible draws, not an implementation bug). The workable target for Metropolis-Hastings under a pseudo-marginal likelihood (Pitt, Silva, Giordani and Kohn, 2012) is variance ≈ 1 ; reaching it by adding particles alone extrapolates to roughly 4.8 million particles and about 34 minutes per likelihood evaluation – making even a short diagnostic chain a multi-day undertaking and this project’s usual protocol (tens of thousands of draws per chain) infeasible outright. A parallel patch of `auxiliary_particle_filter.m` (the same fixed-seed bug, same fix) found the auxiliary particle filter roughly 3.8 times more efficient per particle than the sequential importance filter tested first, which still only halves the problem (about 1.3 million particles, 17 minutes per evaluation) rather than closing it.

Correlated pseudo-marginal MCMC. Deligiannidis, Doucet and Pitt (2018, *Journal of the Royal Statistical Society B* 80(5), 839–870) address exactly this: correlate the random numbers driving the particle filter across MH iterations (a Crank-Nicolson update, $U' = \rho U + \sqrt{1 - \rho^2} Z$) so that the *difference* between two log-likelihood estimates has much lower variance than two independent draws, even though each estimate’s own variance is unchanged. Testing this required rewriting the particle filter to take every source of its randomness – the initial particle cloud, each period’s structural innovations, and (under this project’s default systematic/kitagawa resampling, confirmed to need exactly one uniform draw per period) the resampling draw itself – as an external input rather than calling `randn()`/`rand()` internally, making the filter a pure function of (θ, U) . With that in place, correlation does help substantially: at 5,000 particles the variance of the likelihood *difference* falls from 1620 (independent draws, $\rho = 0$) to 150.8 at $\rho = 0.999$ and 56.1 at $\rho = 0.99999$ – a 30-100-fold reduction. But it plateaus well short of the needed $O(1-10)$: pushing ρ from 0.999 to 0.99999, a perturbation of the underlying randomness of well under 1%, only takes variance from 150.8 to 56.1, not toward the target. The most likely cause is the resampling step itself: systematic resampling maps a single uniform to a *discrete* set of particle indices, so even a near-identical uniform can flip which particles are selected near a weight tie, reintroducing a floor of effectively uncorrelated noise at every one of the 68 resampling events regardless of how close ρ is to 1. The literature’s fix for this specific problem – a resampling scheme designed to vary smoothly in the underlying randomness – is itself a substantial implementation project, not a further parameter adjustment, and was not attempted.

Bottom line. The smooth-ZLB/particle-filter approach was taken as far as is reasonable without a multi-day engineering commitment: the seeding bug is real, documented, and fixed in a reusable project-local patch; the resulting noise is precisely measured rather than assumed; and a genuine attempt at the standard literature fix (correlated pseudo-marginal MCMC) produced real, measured improvement that nonetheless falls short by roughly two orders of magnitude, for an identified and specific reason rather than an unexplained wall. The mode comparison in the preceding paragraphs – not a full posterior – remains the practical ceiling of what this alternative delivers at present, and should be read accordingly: a structural cross-validation of OccBin’s point estimates, not an independent Bayesian estimation of this model.

2.6 Diffuse-prior re-estimation (current round)

Following the reserve-equation fix and prior changes described in Section 1, mode-finding and estimation were restarted from scratch: both the model equations and the `estimated_params` priors changed, so the previous round’s mode file is stale on two independent counts and cannot be reused (a mismatched parameter vector alone causes Dynare to throw “generated using another specification of the model”). Unlike every earlier round, `mcmc_jumping_covariance=hessian` was sufficient throughout – no custom pooled-posterior-draws covariance was needed this time. Table 5 reports the full tuning history.

Stage	mh_jscale	Chain 1	Chain 2	Outcome
Stage 1 (linear KF, <code>prior_variance</code>)	0.20	0.0047	0.0123	b_y mode at exactly 0, non-PD Hessian; b_y prior
Stage 1 rerun (<code>prior_variance</code>)	0.03	0.452	0.443	Healthy; mode/Hessian clean; promoted
Stage 2 (OccBin, <code>hessian</code> , 3k)	0.20	0.102	0.072	Above 0.05 floor but below target; retuned
Stage 2 (<code>hessian</code> , 3k)	0.10	0.280	0.175	Healthy; b_π stable across reruns; promoted
Production (<code>hessian</code> , 50k)	0.10	0.0543	0.0531	Barely above 0.05; Geweke only 2/8, 3/8 conver
Extended diagnostic (<code>hessian</code> , 8k)	0.05	0.390	0.276	Healthy, larger diagnostic used given prior degra
Production (<code>hessian</code>, 50k)	0.05	0.1855	0.1855	Accepted; Geweke 6/8, 5/8 – best conver

Table 5: Tuning history for the diffuse-prior re-estimation round (reserve-equation fix; b_π, b_y diffuse Gamma priors; φ_b prior mean lowered to 0.05). The jscale=0.10 production run’s severe diagnostic-to-production degradation (28.0%/17.5% $\rightarrow$ 5.43%/5.31%, roughly 5 $\times$, worse than any prior round) motivated both a further jscale cut and, given how badly the usual 3,000-draw diagnostic had just mispredicted production behavior, a longer 8,000-draw diagnostic before committing to another full run.

The accepted run’s acceptance ratio is remarkably symmetric between chains (18.55%/18.55%) and its Geweke convergence (6 of 8 and 5 of 8 structural parameters passing at the 15% taper level; full detail in Section 5) matches or exceeds the best result from any earlier round of this estimation exercise. Chain-mean gaps for the four policy parameters, computed directly from each chain’s full retained post-burn-in sample (25,000 draws/chain) in pooled-standard-deviation units, are 0.069 (b_π), 0.203 (b_y), 0.145 (a_i), and 1.416 (φ_b) – b_π , b_y , and a_i all show materially better between-chain agreement than in any previous round (compare Section 2.4’s gaps of 0.13/0.74/0.02 for the same three parameters under the previous, tight-prior round), while φ_b remains the weakest point, consistent with its Geweke failure on chain 1 (Table 9). The Modified-Harmonic-Mean log marginal data density is 1158.40. The completed run took 20h20m.

3 Calibrated Parameters

Table 6 lists every structural parameter held fixed (i.e., *not* in `estimated_params`) for the results reported in this note. Two of these – $\bar{\tau}$ and χ – were recalibrated this round, as described in Section 1; the remainder are unchanged from the baseline `mmt8b0ccBin3` calibration.

4 Posterior Estimates

The headline result of loosening the b_π prior: rather than sitting near the old tight prior’s mean of 2.10, the posterior moves to **3.474** [3.344, 3.604] – decisively above the model’s determinacy threshold ($b_\pi \approx 1.10$ at these b_y, φ_b values; Section 6) and *higher* than the previous round’s tight-prior estimate, not lower. The data want a stronger inflation response than the earlier informative prior allowed; this is not an artifact of prior choice. b_y settles at 0.083 [0.057, 0.106], close to where the old (informative) prior already placed it, and φ_b at 0.069 [0.030, 0.101], modestly above its new prior mean of 0.05. a_i is essentially unchanged from the previous round (0.181 here vs. 0.180 previously) despite its own prior being untouched – a useful consistency check, since everything else in the model moved.

Figure 2 shows prior and posterior densities for the four monetary/fiscal policy parameters – b_π , b_y , a_i , and φ_b – against their new, less informative priors. Posterior densities are kernel density estimates of the pooled post-burn-in draws (50% burn-in per chain, then pooled across both chains for 50,000 draws total); prior densities are the exact analytical densities implied by the hyperparameters in Table 7. Checking the two chains separately (using each chain’s full retained post-burn-in sample, not just Geweke’s early/late sub-windows), the chain-mean gaps are 0.069, 0.203, 0.145, and 1.416 pooled standard deviations for b_π , b_y , a_i , φ_b respectively – b_π , b_y , and a_i all show better between-chain agreement than in any earlier round of this estimation exercise (compare 0.13, 0.74, 0.02 under the previous, tight-prior round’s best jumping-covariance choice), while φ_b is the new round’s weak point.

The posterior for b_π is tight and comfortably above unity even under a prior with mean 1 and support well past 3, confirming the Taylor principle is satisfied on the strength of the data rather than the prior. The COVID labor-supply shock’s standard deviation σ_L again has a 90% HPD interval well away from zero ([0.062, 0.076]), against a deliberately loose prior (`InvGamma(0.15,0.075)`) chosen so that a single-quarter outlier would not be crowded out by measurement error absorbing it instead.

For reference, Table 8 reproduces the previous (tight-prior) round’s posterior estimates, under the pre-fix reserve equation and the original b_π, b_y, φ_b priors described in Section 2.3.

5 Geweke (1992) Convergence Diagnostics

Table 9 reports Geweke’s convergence p -values (15% taper, χ^2 test for equality of early- vs. late-chain means) for the eight estimated structural parameters, computed separately for each chain, for the accepted current-round (`hessian, mh_jscale=0.05`) run. A parameter is classified as converged if $p > 0.05$.

This is the best convergence result of any round in this estimation exercise: chain 1 passes 6 of 8 structural parameters, chain 2 passes 5 of 8 – versus 6/8 and 4/8 for the previous round’s best (`posterior-cov`) result, and markedly better than the same current round’s own first production attempt at `mh_jscale=0.10` (2/8 and 3/8; see Table 5), which motivated the further

Symbol	Description	Value
<i>Household / preferences</i>		
β	Discount factor	0.9967
χ	Shopping-time curvature	0.5 (recalibrated from 1.0)
ψ	Inverse Frisch elasticity	1.3
z	Long-run labor productivity	1.3
<i>Banking sector</i>		
ν	Deposit CES elasticity	0.25
x_n	Loan portfolio share	0.95
$x_h = 1 - x_n$	Deposit-labor portfolio share	0.05
ϕ_n	Reserve management cost	0.0005
x_a	Deposit-technology level (steady state)	5.2528
<i>Price setting</i>		
θ	Goods demand elasticity	6.0
ϕ_p	Rotemberg price-adjustment cost	50
<i>Fiscal / debt</i>		
g	Steady-state govt. spending (share of output)	0.20
$\bar{\tau}$	Steady-state tax target	0.14 (recalibrated from 0.20)
$g - \bar{\tau}$	Implied steady-state deficit/GDP	6% (from $\approx 0\%$)
b_0	Steady-state debt target	1.0
ω	Fraction of bonds held by the central bank	0.0
<i>Nominal anchors</i>		
π^{ss}	Reference gross quarterly inflation (2% annual)	1.004963
$r^{\ell,ss}$	Reference gross quarterly loan rate (Fisher: π^{ss}/β)	1.008290
$\underline{r}^n$	ZLB floor (gross quarterly rate)	1.0
<i>Shock persistence (fixed, not estimated)</i>		
ρ_a	TFP persistence	0.95
ρ_{xa}	Deposit-technology persistence	0.95
ρ_{va}	Preference-shock persistence	0.0
ρ_g	Govt. spending persistence	0.0
ρ_τ	Tax-rule persistence	0.5

Table 6: Calibrated (non-estimated) parameters for the results in this note. π^{ss} and $r^{\ell,ss}$ are reference values used only in the tax-rule/debt-issuance block (Section 1) and the Taylor-rule normalization; the model’s actual *endogenous* non-ZLB steady state, reported in Table 10, need not equal them exactly. ρ_a, ρ_{xa} are fixed rather than estimated because freeing them previously drove ρ_a to a near-unit-root value and produced a non-positive-definite Hessian at the mode (Section 2.2).

Parameter	Prior mean	Post. mean	90% HPD	Prior
v^a (preference scale)	2.500	2.890	[2.055, 3.482]	Gamma(2.50, 0.75)
η (labor preference)	1.530	1.136	[0.922, 1.326]	Gamma(1.53, 0.40)
b_π (inflation response)	1.000	3.474	[3.344, 3.604]	Gamma(1.00, 1.00)
b_y (output response)	1.000	0.083	[0.057, 0.106]	Gamma(1.00, 0.70)
a_i (policy-rate level weight)	0.490	0.181	[0.169, 0.195]	Beta(0.49, 0.10)
b_r (MP shock persistence)	0.360	0.262	[0.171, 0.347]	Beta(0.36, 0.10)
ρ_z (productivity persist.)	0.450	0.443	[0.292, 0.549]	Beta(0.45, 0.10)
φ_b (fiscal feedback)	0.050	0.069	[0.030, 0.101]	Beta(0.05, 0.05)
<i>Standard deviation of shocks</i>				
σ_a	0.050	0.017	[0.016, 0.018]	InvGamma(0.05, 0.025)
σ_z	0.050	0.023	[0.018, 0.028]	InvGamma(0.05, 0.025)
σ_{va}	0.050	0.039	[0.026, 0.050]	InvGamma(0.05, 0.025)
σ_{xa}	0.050	0.036	[0.032, 0.041]	InvGamma(0.05, 0.025)
σ_g	0.050	0.030	[0.025, 0.036]	InvGamma(0.05, 0.025)
σ_r	0.050	0.024	[0.021, 0.026]	InvGamma(0.05, 0.025)
σ_τ	0.050	0.031	[0.024, 0.036]	InvGamma(0.05, 0.025)
σ_L (COVID labor-supply)	0.150	0.069	[0.062, 0.076]	InvGamma(0.15, 0.075)
$\sigma_{me,y}$	0.050	0.021	[0.018, 0.023]	InvGamma(0.05, 0.025)
$\sigma_{me,d}$	0.050	0.018	[0.015, 0.022]	InvGamma(0.05, 0.025)
$\sigma_{me,\pi}$	0.050	0.014	[0.013, 0.014]	InvGamma(0.05, 0.025)
$\sigma_{me,r}$	0.050	0.010	[0.009, 0.012]	InvGamma(0.05, 0.025)
σ_{me,r^n}	0.010	0.003	[0.002, 0.003]	InvGamma(0.01, 0.005)
$\sigma_{me,\tau}$	0.050	0.020	[0.017, 0.024]	InvGamma(0.05, 0.025)
$\sigma_{me,bT}$	0.050	0.022	[0.019, 0.025]	InvGamma(0.05, 0.025)

Table 7: Posterior estimates, current (diffuse-prior) round: 2 chains $\times$ 50,000 draws, `mh_jscale=0.05`, `hessian` jumping covariance, true OccBin ZLB likelihood, reserve-equation fix (Section 1), FYGFDPUN-based debt observable, $\bar{\tau} = 0.14$, $\chi = 0.5$. ρ_a, ρ_{xa} remain fixed at 0.95 (see Section 2.2).

jscale cut. b_π now passes on *both* chains, despite its posterior having moved substantially under the diffuse prior – the convergence result is not an artifact of the prior pinning the estimate near a fixed value. a_i and ρ_z fail on one chain each (having each passed on both chains in the previous round), and b_r fails on both chains, roughly matching the previous round’s pattern for that parameter. η ’s chain 2 failure ($p = 0.008$) is the one parameter that converged in the previous round but does not fully here; its posterior HPD [0.922, 1.326] remains a reasonably tight, economically sensible range regardless. As in every previous round, no single jumping-covariance or jscale choice achieves full 8/8 convergence on both chains simultaneously – b_r and (on one chain) a_i, ρ_z remain the persistent stragglers here, analogous to b_y ’s role in the previous round.

6 Determinacy and Expectational Stability

Determinacy (Blanchard and Kahn, 1980) and E-stability (Evans and Honkapohja, 2001, Proposition 10.3) are evaluated on the RELAX-regime linear model – i.e., the model with $r_t^n = r_t^{n,notional}$ imposed unconditionally, with no `occbin_constraints` block and no ZLB floor – using a companion file, `mmt8bOccBin3_covid_bT_baa_bkonly.mod`, that strips the `estimated_params/estimation(...)`

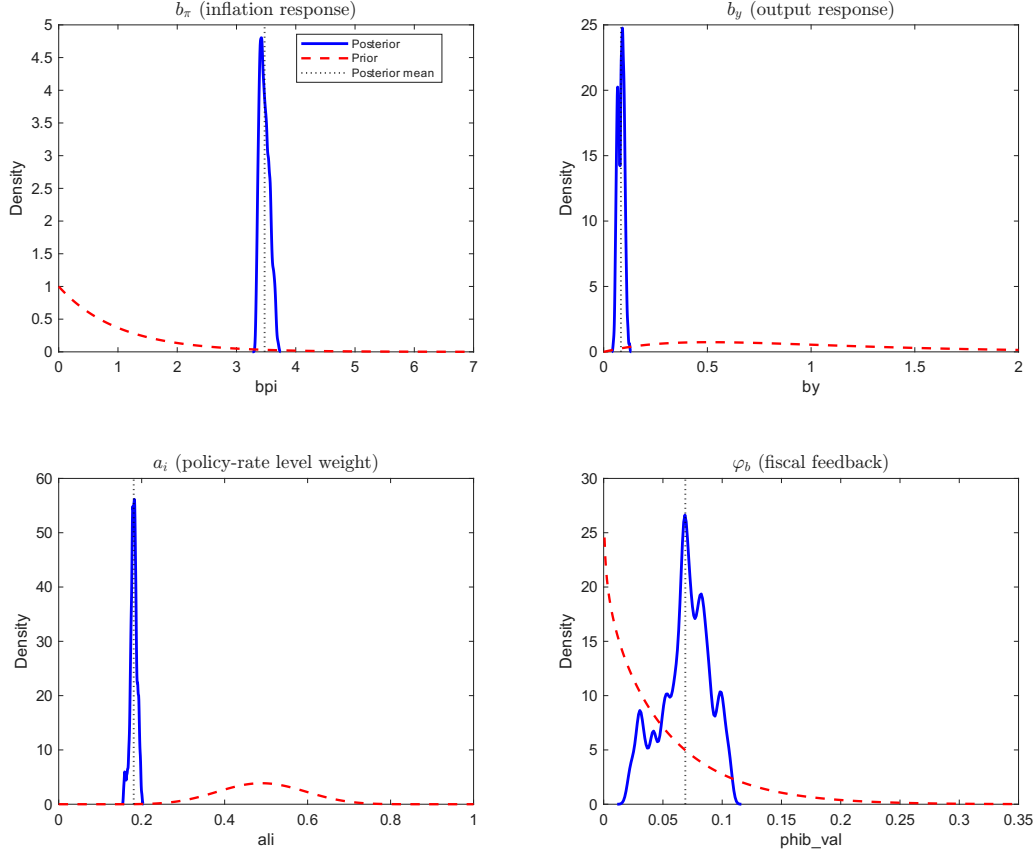


Figure 2: Prior (dashed red) and posterior (solid blue) densities for the policy parameters b_π , b_y , a_i , and φ_b , current (diffuse-prior) round. Dotted vertical lines mark the posterior mean.

machinery but otherwise retains every shock from the estimated model, including ε_t^L . Purely contemporaneous iid shocks with no own law of motion do not enter the endogenous-variable Jacobian and therefore cannot affect the Blanchard-Kahn eigenvalue count or the E-stability spectral radius; what isolates the non-ZLB equilibrium is dropping the OccBin regime-switching equation, not which shocks are declared.

All other parameters are held at their posterior means from Table 7 (current, diffuse-prior round). The reserve equation fix (Section 1) leaves this analysis mechanically unaffected – n_t does not enter the RELAX-regime Jacobian’s forward-looking block – and was verified separately to leave the Blanchard-Kahn eigenvalue count (4 forward-looking variables) and steady state unchanged. Figure 3 sweeps $(b_\pi, b_y) \in [0, 5] \times [0, 3]$ on a 60×60 grid – the b_π range extended from the earlier rounds’ $[0, 3]$ specifically so the current round’s posterior mean, which the diffuse prior moved to $b_\pi = 3.474$, actually falls inside the plotted region rather than off its edge. Of the 3,600 grid points, 2,857 are simultaneously determinate and E-stable. The determinate and E-stable classifications were verified to coincide exactly, cell-for-cell, with zero mismatches across all 3,600 grid points, as expected from Proposition 10.3 for this class of model. Because the two conditions are literally identical here, they are combined into a single two-region figure rather than shown as separate determinacy and E-stability panels. At the posterior mean $(b_\pi, b_y) = (3.474, 0.083)$ the model is determinate and E-stable – comfortably inside the determinate region, since the

Parameter	Prior mean	Post. mean	90% HPD	Prior
v^a	2.500	2.776	[1.526, 3.830]	Gamma(2.50, 0.75)
η	1.530	0.988	[0.666, 1.266]	Gamma(1.53, 0.40)
b_π	2.100	2.169	[2.015, 2.315]	Normal(2.10, 0.15)
b_y	0.280	0.135	[0.068, 0.188]	Gamma(0.28, 0.10)
a_i	0.490	0.180	[0.161, 0.199]	Beta(0.49, 0.10)
b_r	0.360	0.232	[0.151, 0.333]	Beta(0.36, 0.10)
ρ_z	0.450	0.562	[0.410, 0.720]	Beta(0.45, 0.10)
φ_b	0.100	0.094	[0.066, 0.124]	Beta(0.10, 0.05)

Table 8: Previous round’s posterior estimates (structural parameters only), reproduced for comparison: pre-fix reserve equation, tight b_π, b_y, φ_b priors, custom pooled-posterior jumping covariance, `mh_jscale=0.25`.

Parameter	Chain 1 p	Converged?	Chain 2 p	Converged?
v^a	0.350	Yes	0.558	Yes
η	0.088	Yes	0.008	No
b_π	0.062	Yes	0.089	Yes
b_y	0.281	Yes	0.070	Yes
a_i	0.024	No	0.563	Yes
b_r	0.001	No	0.001	No
ρ_z	0.564	Yes	0.000	No
φ_b	0.226	Yes	0.081	Yes

Table 9: Geweke convergence test p -values, 15% taper, by chain (current diffuse-prior round, `hessian` covariance, `mh_jscale=0.05`, $\bar{\tau} = 0.14$, $\chi = 0.5$).

grid’s own determinacy threshold along the $b_y = 0.083$ line is $b_\pi \approx 1.10$ and the posterior mean sits more than three times that value.

As a robustness check on the fiscal feedback parameter φ_b , determinacy and E-stability were re-evaluated at $\varphi_b \in \{0.0301, 0.0689, 0.1005\}$ (the current run’s 90% HPD lower bound, posterior mean, and HPD upper bound), holding (b_π, b_y) at their posterior means:

φ_b	Determinate?	Spectral radius	E-stable?
0.0301 (HPD lower)	Yes	0.9948	Yes
0.0689 (posterior mean)	Yes	0.9837	Yes
0.1005 (HPD upper)	Yes	0.9744	Yes

Stability holds at all three points, with the spectral radius falling (i.e., stability strengthening) as φ_b rises across this range, as in every previous round. The spectral radii are somewhat closer to the instability boundary (1.0) than in the previous round (which ranged 0.9674–0.9844 across its φ_b HPD), consistent with this round’s lower φ_b HPD range sitting closer to zero, but stability is not in question at any of the three points.

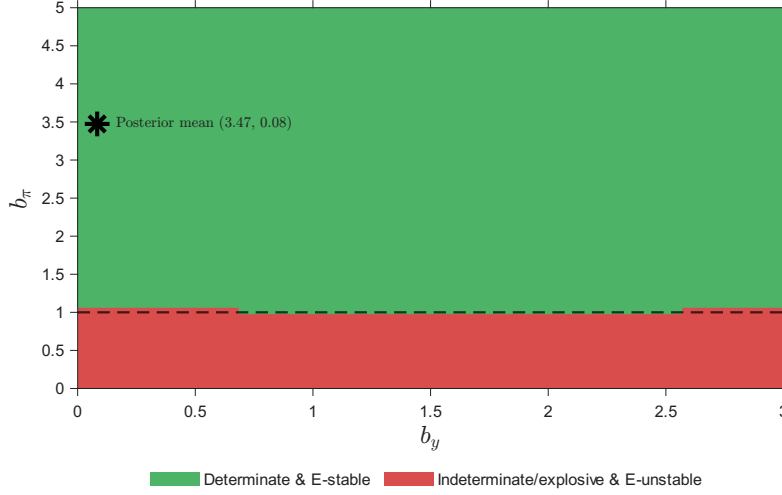


Figure 3: Combined determinacy/E-stability region over $(b_\pi, b_y) \in [0, 5] \times [0, 3]$, all other parameters at posterior means (current, diffuse-prior round). Green: determinate *and* E-stable (the two conditions coincide exactly on this grid, with 0/3,600 mismatched cells). Red: indeterminate or explosive, and E-unstable. The star marks the posterior mean $(b_\pi, b_y) = (3.474, 0.083)$, now visible within the plotted range.

7 Steady-State Annualized Rates

Table 10 reports the model’s steady-state gross quarterly and annualized rates evaluated at the full posterior mean parameter vector (Table 7, current diffuse-prior round), obtained by resolving the non-ZLB (RELAX) steady state with `resol()` after loading the posterior means – necessary because v^a, η (which do enter the steady state, unlike $b_\pi, b_y, a_i, b_r, \rho_z, \varphi_b$, which affect only the model’s log-linear dynamics around it) moved from the previous round. As a consistency check, the steady-state real rate satisfies the Euler equation exactly: $r^\ell/\pi = 1/\beta = 1.003311$.

Symbol	Description	Gross quarterly	Annualized (%)
π	Inflation	1.008964	3.63
r^ℓ	Loan rate	1.012304	5.01
r^d	Deposit rate	1.004333	1.74
r^n	Policy rate (interest on reserves)	1.002214	0.89

Table 10: Steady-state gross quarterly rates and simple annualized percentages $((\text{rate}^4 - 1) \times 100)$ at the posterior mean parameter vector, non-ZLB (RELAX) regime, $\bar{\tau} = 0.14$, $\chi = 0.5$, current (diffuse-prior) round.

These are close to, but not identical to, the previous round’s steady state (3.79% vs. 3.63% inflation, 1.86% vs. 1.74% deposit rate) – a modest shift consistent with v^a and η ’s posterior means moving somewhat (Table 7 vs. Table 8) while $\bar{\tau}, \chi, \beta$ and every other steady-state-relevant calibrated parameter are unchanged; the reserve-equation fix itself leaves the steady state exactly unchanged (verified directly – see Section 1). Both qualitative features noted in the previous round persist. First, steady-state annual inflation (3.63%) remains well above the original ($\bar{\tau} = 0.20$, $\chi = 1.0$) calibration’s 1.10%, consistent with the larger calibrated deficit ($\bar{\tau}$ lowered

from 0.20 to 0.14) dominating the lower χ 's inflation-reducing effect. Second, the steady-state deposit rate r^d remains positive (1.74% annualized), as in the previous round, in contrast to the fractionally negative values seen under the original calibration. The discount factor $\beta = 0.9967$ was originally chosen so that the loan rate would clear a 2% inflation target while keeping the deposit rate above the ZLB; under the recalibrated $\bar{\tau}, \chi$ that margin remains comfortably positive.

8 Counterfactual Analysis: Debt and Monetary Policy in the Post-COVID Inflation Episode

Beyond the unconditional posterior described above, two exercises ask a conditional-on-the-data question: given the smoothed history implied by this model and the posterior mean of the production round with σ_{me,r^n} fixed at 0.0015 (the round whose regime path recovers the true ZLB dates; Section 2.5), which shocks actually account for the elevated inflation observed over 2021Q2–2023Q2 (the period this note's abstract refers to as “elevated inflation”)? Section 8.1 describes the shared methodology; Sections 8.2 and 8.3 report the debt/fiscal and monetary-policy exercises, respectively.

8.1 Methodology: historical decomposition and shock-off counterfactuals

Both exercises use Dynare's `calib.smoother` command – the Kalman smoother evaluated once at fixed (posterior-mean) parameters, without any further MCMC – applied to the same estimation dataset (`fred_data_debt_baa.mat`) used throughout this note, followed by `shock.decomposition`. The parameters are the posterior means of the production round with σ_{me,r^n} fixed at 0.0015 ($b_\pi = 3.355$, $b_y = 0.083$, $a_i = 0.184$, $b_r = 0.252$, $\varphi_b = 0.091$). The same exercises were also run at the posterior means of the earlier round in which σ_{me,r^n} was estimated (Table 7; $b_\pi = 3.474$). The two rounds give nearly identical results – the 2021Q2–2023Q2 average inflation deviations and counterfactuals agree to within about 4%, and the only visible difference is the size of the debt counterfactual (about 27 against 32 percentage points of GDP) – so the tables and figures below report the fixed- σ_{me,r^n} round, and Table 14 gives the comparison. For a variable x_t observed or implied by the model, this yields an exactly additive decomposition of its smoothed deviation from steady state,

$$\hat{x}_t - \bar{x} = \sum_{k=1}^{15} \phi_{k,t} + \iota_t,$$

where $\phi_{k,t}$ is the k -th shock's own contribution at t (computed by simulating the linear state-space system with only shock k 's smoothed innovations active) and ι_t is the contribution of the smoothed initial condition; summing all sixteen terms reproduces $\hat{x}_t - \bar{x}$ to numerical precision, which was verified directly. Because this decomposition is linear and exactly additive, a shock-off counterfactual does not require re-simulating the model: it is simply the actual smoothed path with one or more shocks' contributions subtracted,

$$x_t^{\text{cf}} = \bar{x} + \left(\hat{x}_t - \bar{x} \right) - \sum_{k \in \mathcal{K}} \phi_{k,t},$$

for the shock set $\mathcal{K}$ being switched off.

Both exercises use the *linear*, non-OccBin Kalman smoother – the same approximation used for this note’s own Stage-1 mode-finding (Section 2.2) – rather than the true OccBin piecewise-linear ZLB smoother, which (Section 2.5) does not reach a fixed point for this model. This matters wherever the floor binds. The target window, 2021Q2–2023Q2, is *not* entirely past liftoff: the regime path verified in Section 2.5 has the floor binding through 2022Q2 (the observed-rate episode in Section 1 ends 2021Q4, and the linear notional-rate check 2022Q1), so roughly the first five of the window’s nine quarters (2021Q2–2022Q2) are at or near the floor. In those quarters the linear smoother is infeasible by construction – its policy rate averages 0.9994 (gross quarterly), below the floor of 1 – and, under a true floor, ε_t^r is not point-identified (any shock pushing the notional rate below 1 is consistent with the data), whereas the linear smoother picks the single value matching the observed rate. Section 8.4 therefore splits the window into binding and post-liftoff quarters (2022Q3–2023Q2) and reports both. The COVID-quarter numbers reported for contrast should be read with the same caveat.

8.2 Do debt-moving shocks explain the inflation spike?

Debt b_t (and hence bT_t) has no exogenous shock of its own in this model – it is an accumulated state, driven by the government budget identity (Section 1) – so the relevant counterfactual switches off the two shocks that *directly* enter that block, the tax-rule shock ε_t^r and the government-spending shock $\varepsilon_{g,t}$ (Table 1), and asks how bT_t and π_t would have evolved without them.

Window	Variable	Avg. actual dev. from SS	Avg. counterfactual dev. from SS
2021Q2–2023Q2	π_t	0.00290	0.00295
2021Q2–2023Q2	bT_t	0.1585	−0.1110
2020Q1–2020Q4 (COVID)	π_t	−0.00112	−0.00017

Table 11: Actual vs. counterfactual ($\varepsilon_t^r, \varepsilon_{g,t}$ shut off) average deviations from steady state. π_t in gross quarterly units, bT_t in debt/GDP ratio units.

The debt panel confirms the experiment is doing what it should: without $\varepsilon_t^r, \varepsilon_{g,t}$, debt-to-GDP would average roughly 27 percentage points lower over 2021Q2–2023Q2 (0.71–0.73 rather than the actual 0.98–1.01; about 32 points at the posterior mean of the estimated- σ_{me,r^n} round) – these two shocks are indeed the model’s debt-moving shocks. But the inflation panel shows essentially no corresponding gap (0.00290 actual vs. 0.00295 counterfactual, i.e. debt-related shocks account for about 0% of the average inflation deviation over that window; 0.00302 vs. 0.00307 in the estimated- σ_{me,r^n} round); the two paths are visually indistinguishable from 2021Q2 through 2023Q2 in Figure 4. The one episode where the two paths diverge visibly is the COVID quarters themselves (2020Q2–Q3), where the deficit shock’s direct demand effect does move inflation (a −0.00095 average gap over 2020Q1–Q4). In this estimated model, debt accumulation and the 2021–2023 inflation episode are essentially decoupled: the debt buildup is fiscal-shock-driven as expected, but the subsequent inflation surge is not.

8.3 Was monetary policy behind the curve?

The second exercise switches off only the monetary-policy shock ε_t^r (Table 1) and compares the actual smoothed π_t and r_t^n to what the estimated policy rule alone – with $\varepsilon_t^r \equiv 0$, i.e.

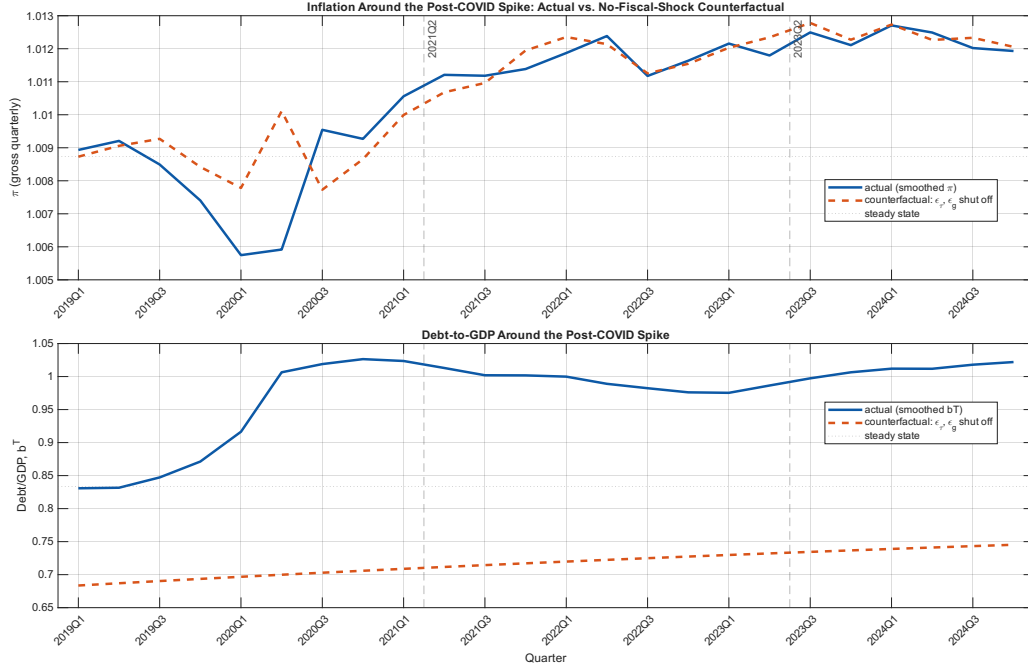


Figure 4: Actual (smoothed) vs. counterfactual π_t (top) and bT_t (bottom) with $\varepsilon_t^r, \varepsilon_{g,t}$ shut off, 2019Q1–2024Q4. Posterior mean (fixed- σ_{me,r^n} round), linear (non-OccBin) Kalman smoother.

$z_t^r = b_r z_{t-1}^r$ decaying from its initial condition rather than being perturbed each period – would have produced given the rest of the smoothed system.

Window	Variable	Avg. actual	Avg. counterfactual (ε_t^r shut off)
2021Q2–2023Q2	π_t dev. from SS	0.00290	−0.00221
2021Q2–2023Q2	r_t^n level	1.00295	0.99794
2020Q1–2020Q4 (COVID)	π_t dev. from SS	−0.00112	0.00014

Table 12: Actual vs. counterfactual (ε_t^r shut off) average deviations/levels. π_t, r_t^n in gross quarterly units.

Without ε_t^r , inflation over 2021Q2–2023Q2 would have averaged *below* steady state (−0.00221) instead of above it (+0.00290 actual) – a 0.00512 average gap, matching the ε_t^r row of the shock decomposition exactly, since the two constructions are algebraically equivalent (at the estimated- σ_{me,r^n} posterior mean: −0.00215 against +0.00302, a 0.00517 gap). The policy-rate counterfactual shows the actual rate running about 2 percentage points (annualized) above what the estimated rule alone would have set. Pulling the smoothed shock series directly confirms the sign and persistence of this: ε_t^r is negative in every quarter from 2021Q2 through 2023Q2 (mean −0.0089), i.e. the smoother reads monetary policy as systematically looser than the estimated rule ($b_\pi = 3.355$) would prescribe given realized π_t, y_t throughout the episode. This is the mirror image of the early COVID quarters, where ε_t^r is positive in 2020Q1–Q2 (mean +0.0015 over 2020Q1–2020Q4) and the counterfactual sign flips accordingly (inflation would have been *higher*, not lower, without the shock). Note the sign of the policy-rate counterfactual: although a negative ε_t^r lowers the rule’s rate directly, the actual rate sits *above* its shock-off counterfac-

tual (1.00295 against 0.99794). The shock raises inflation by 0.00512 on average, and the rule’s response to that extra inflation ($b_\pi \times 0.00512 \approx 0.017$) outweighs the direct effect of the shock, so its net effect on the policy rate is positive (about +0.005, or 2 percentage points annualized). Both inflation and, on net, the policy rate thus move opposite to the shock’s sign – the shock’s net effect once propagated through the full general-equilibrium system (including the banking block’s rate spreads), not merely its direct, partial-equilibrium effect on the policy rule alone.

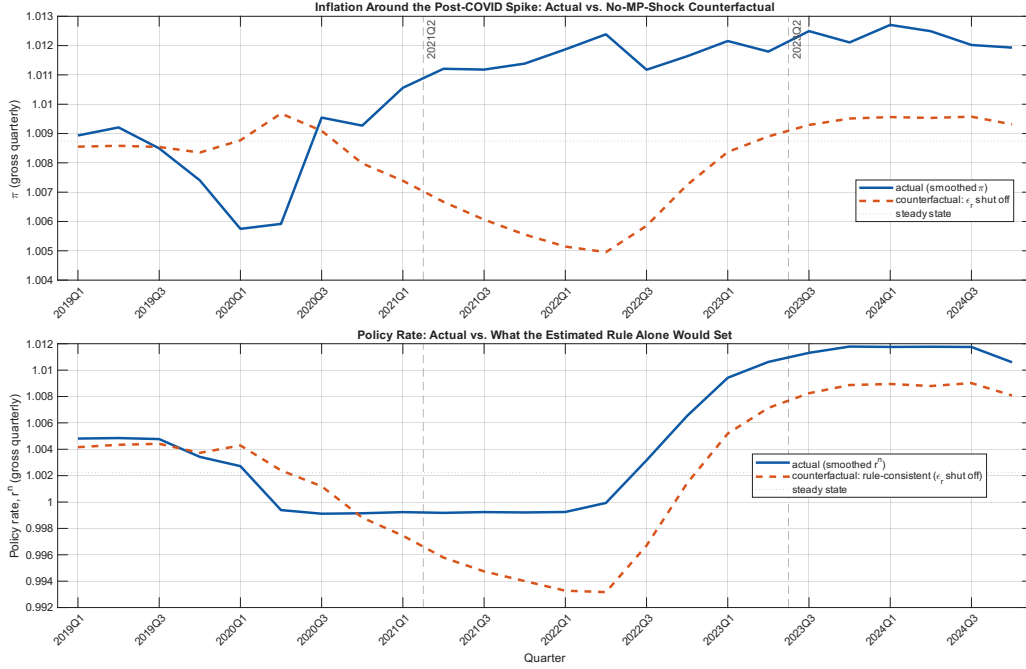


Figure 5: Actual (smoothed) vs. counterfactual π_t (top) and r_t^n (bottom, actual vs. what the estimated rule alone would set) with ϵ_t^r shut off, 2019Q1–2024Q4. Posterior mean (fixed- σ_{me,r^n} round), linear (non-OccBin) Kalman smoother.

8.4 Sensitivity to the ZLB quarters

Because contributions are linear in the smoothed innovations, each shock’s average contribution over 2021Q2–2023Q2 can be split exactly by *when* the underlying innovations were realized (simulating each shock once per block of innovation dates; the pieces sum to the decomposition of Section 8.1 to five decimals). An earlier version of this table used three blocks, treating 2020Q1–2021Q1 as the ZLB quarters and everything from 2021Q2 on as “in-window”; that definition put the binding quarters 2021Q2–2022Q2 inside the window and so could not test contamination from them. The split below separates them:

Of the monetary-policy shock’s 0.00512 contribution to inflation, about 70% (0.00359) comes from innovations realized in the binding quarters 2021Q2–2022Q2, 25% (0.00128) from post-liftoff innovations, and only 5% (0.00024) from 2020Q1–2021Q1 (68%, 27% and 5% at the estimated- σ_{me,r^n} posterior mean). The earlier conclusion that the result “does not depend on the smoothed shocks of the ZLB quarters” therefore held only for the narrower 2020Q1–2021Q1 definition: the window’s largest contributions *do* come from quarters where the floor binds. The fiscal shocks’ contribution to inflation remains essentially zero from every origin, and the

Variable	Shock	Pre-2020	2020Q1–21Q1	2021Q2–22Q2 (binding)	2022Q3+ (post-liftoff)	Total
π_t	ε_t^r	−0.00000	0.00024	0.00359	0.00128	0.00512
π_t	ε_t^r	−0.00000	−0.00000	0.00000	0.00000	0.00000
π_t	$\varepsilon_{g,t}$	0.00000	0.00004	−0.00002	−0.00007	−0.00005
bT_t	$\varepsilon_{g,t}$	0.16707	0.12952	0.00369	0.00250	0.30278
bT_t	ε_t^r	−0.02246	0.00799	−0.01884	0.00010	−0.03321

Table 13: Average contribution over 2021Q2–2023Q2, split by the dates of the innovations generating it (fixed- σ_{me,r^n} round). Columns 3–6 nest the earlier three-block split (old “in-window” = columns 5 + 6).

debt contribution is still dominated by pre-window (chiefly COVID-era) deficits, so the fiscal conclusion is unaffected.

The sharper question is whether the monetary-policy result survives on the post-liftoff quarters alone, where the linear smoother is not infeasible. Table 14 reports averages by sub-window for the posterior mean of the production round with σ_{me,r^n} fixed at 0.0015 (the round used throughout this section; its regime path recovers the true ZLB dates, Section 2.5) and, for comparison, for the earlier round in which σ_{me,r^n} was estimated (Table 7).

Round	Window	Actual π dev.	ε^r contrib.	Cf. (ε^r off)	Mean ε^r	Mean r^n
σ_{me,r^n} fixed	2021Q2–2023Q2	0.00290	0.00512	−0.00221	−0.0089	1.0030
σ_{me,r^n} fixed	2021Q2–2022Q2 (binding)	0.00286	0.00593	−0.00307	−0.0113	0.9994
σ_{me,r^n} fixed	2022Q3–2023Q2 (post-liftoff)	0.00295	0.00410	−0.00115	−0.0058	1.0074
σ_{me,r^n} estimated	2021Q2–2023Q2	0.00302	0.00517	−0.00215	−0.0094	1.0029
σ_{me,r^n} estimated	2021Q2–2022Q2 (binding)	0.00292	0.00581	−0.00289	−0.0115	0.9995
σ_{me,r^n} estimated	2022Q3–2023Q2 (post-liftoff)	0.00315	0.00438	−0.00123	−0.0068	1.0072

Table 14: Monetary-policy counterfactual by sub-window, linear smoother at the posterior mean of each round. π and r^n in gross quarterly units; ε^r contribution = actual minus counterfactual deviation (rows may differ by rounding).

It does. On the post-liftoff quarters alone, shutting off ε_t^r still moves average inflation from +0.00295 to −0.00115 (fixed- σ_{me,r^n} round) and from +0.00315 to −0.00123 (estimated- σ_{me,r^n} round): the shock’s contribution (0.00410, 0.00438) exceeds the entire actual deviation in both rounds, and the smoothed shock remains negative (mean −0.0058, −0.0068). The binding-quarter figures are larger (0.00593, 0.00581; mean shock about −0.0113) but are the ones the linear smoother cannot be trusted on: the smoothed policy rate there averages 0.9994, below the floor. They describe what the linear model needs in order to match the observed rate, not an estimate of what policy contributed.

The two rounds are not noticeably different for this section’s questions: over 2021Q2–2023Q2, inflation averages 0.00290 against a debt-shocks-off counterfactual of 0.00295 (estimated round: 0.00302 against 0.00307), and the ε_t^r -off counterfactual is −0.00221 (−0.00215); the COVID-quarter averages and policy-rate counterfactual (1.00295 against 0.99794, versus 1.00293 against 0.99790) are likewise nearly identical. The one visible difference is the size of the debt counterfactual, about 27 percentage points of GDP lower without $\varepsilon_t^r, \varepsilon_{g,t}$ against about 32 – a 16% difference in a diagnostic whose message (these shocks drive debt, not inflation) is the same in

both. The policy-rule coefficients behind these results are also close across likelihood methods: Section 2.5 reports that an independent smooth-floor, particle-filter mode search lands within roughly 1–4% of OccBin’s mode on every structural parameter. That cross-validation concerns parameters, not the smoother, and does not supply a ZLB-consistent smoothed shock path.

8.5 Summary

Taken together, these two exercises point to monetary policy, not fiscal policy or debt accumulation, as the model’s own account of what drove the 2021Q2–2023Q2 inflation episode: the debt-moving shocks $(\varepsilon_t^T, \varepsilon_{g,t})$ explain essentially none of the average inflation deviation over that window despite clearly driving the contemporaneous debt buildup, while the monetary-policy shock (ε_t^r) alone accounts for more than the entire average deviation (0.00512 against an actual deviation of 0.00290 – other shocks net out mildly negative over the same window). The monetary-policy result survives restriction to the post-liftoff quarters 2022Q3–2023Q2 (contribution 0.00410 against an actual deviation of 0.00295) and is essentially unchanged at the posterior mean of the earlier round with σ_{me,r^n} estimated (0.00517 against 0.00302 for the full window; 0.00438 against 0.00315 post-liftoff), but about 70% of the full-window contribution comes from innovations in the binding quarters 2021Q2–2022Q2, where the linear smoother is infeasible (Section 8.4); the full-window magnitude should not be read as precise. Both findings are conditional on the linear-smoother approximation (Section 8.1) and on the posterior mean; they are not restated as unconditional posterior results and would need to be rechecked against a full OccBin-consistent smoother if and when the fixed-point issue of Section 2.5 is resolved. The smooth-floor cross-validation reported there supports the parameters behind these counterfactuals but does not remove that dependence.